

UHBC – L3 Computer Science – Mobile Applications

Faculty of Sciences | Department of Computer Science – L3
Compilation of Course Questions and Answers
Exams: 2021-2022 | 2022-2023 | 2023-2024 | 2024-2025

EXAM 1 — Resit 2021-2022 (6 pts)

Q1. Explain the integration step in the mobile application development lifecycle (1.5 pts)

The integration (development) step consists of bringing the application to life while respecting the design guidelines of the various operating systems. It includes:

- **Zoning:** Roughly sketching a web page using blocks, showing the placement of content areas and main features.
- **Wireframes:** Using the zonings to detail each block by introducing content (images, text, videos, etc.).
- **Mockup:** Transforming a wireframe into an interactive page.
- **Prototype:** Validating the technologies needed to ensure the project works correctly.

Q2. Illustrate the durable and transient states of a mobile application (1.5 pts)

- **Durable states:** Resumed, Paused, Stopped.
- **Transient states:** Created, Started.
- **Cycle:** onCreate() → onStart() → onResume() [Resumed] → onPause() [Paused] → onStop() [Stopped] → onDestroy() / onRestart() → onStart()

Q3. Give the advantages and disadvantages of Android (1.5 pts)

Advantages:

- Robust OS Kernel.
- Innovative library.
- Ease of development.
- Fast execution.

Disadvantages:

- Performance considerations requiring powerful and fast devices.
- Difficulty of integration for vendors and strong dependency on Google.

Q4. Define a mobile application (1.5 pts)

A mobile application is application software that can be downloaded for free or for a fee, and run from the operating system of a mobile phone, smartphone, or digital tablet.

EXAM 2 — Resit 2022-2023 (6 pts)

Q1. List the data storage locations in an Android application (1.5 pts)

Local storage:

- Files in internal storage or SD card.
- Shared Preferences (SharedPreferences).
- SQLite database.

External storage (Internet / Cloud):

- Data on a server (e.g., MySQL).

- Firebase.

Q2. Durable and transient states

(Same as Exam 1 — see above.)

Q3. Advantages and disadvantages of Android

(Same as Exam 1 — see above.)

Q4. On what basis does the choice of storage depend? (1.5 pts)

a. Data structure:

- Structured data → Relational DBMS (SQLite, MySQL...)
- Semi-structured data → JSON format
- Variables → Shared Preferences
- Unstructured data → Files

b. Data confidentiality:

- Confidential data (e.g., banking app) → private secure server with encryption.
- Non-confidential data → public Cloud service such as Firebase.

EXAM 3 — Mid-term 2023-2024 (6 pts)

Q1. List four characteristics of mobile terminals

- Mobility.
- Use of battery as energy source.
- Limited processing capacity.
- Limited storage capacity.
- More vulnerable to attacks (security).

Q2. Define a service in a mobile application

A service is:

- A process.
- With no graphical interface.
- Running in the background.
- Can be internal or external.

Q3. Data storage locations

(Same as Exam 2 — see above.)

Q4. Durable and transient states

(Same as Exam 1 — see above.)

EXAM 4 — Resit 2024-2025 (6 pts)

Q1. Give the diagram of the mobile application lifecycle (3 pts)

Steps of the mobile application lifecycle:

- **Definition:** Identify needs, user type, added value, usage environment.

- **Design:** Zoning, wireframes, mockup, prototype.
- **Development (Integration):** Code the application while respecting guidelines.
- **Testing:** Functional, load, and ergonomics tests.
- **Deployment:** Publication on app stores.
- **Tracking & User Feedback:** Collect user reviews, update.

Q2. In which phase should the following tasks be completed? (2.5 pts)

Task	Phase
Choosing the type of user	Definition
Validating usability/ergonomics	Testing
Zoning	Design
User review system	Tracking & User Feedback
Setting up a prototype	Design

EXAM 5 — Makeup 2024-2025 (5 pts)

Q1. Four characteristics of mobile terminals

(Same as Exam 3 — see above.)

Q2. Define a service

(Same as Exam 3 — see above.)

Q3. In which phase should the following tasks be completed?

Task	Phase
Mockup	Design
Environment in which the app will be used	Definition
Added value vs. existing apps	Definition
Evaluating the application's load capacity	Testing

EXAM 6 — Mid-term 2024-2025 (5 pts)

Q1. List the criteria for judging that a program is a mobile application (1.5 pts)

- Software-based.
- Application-level (interactive).
- Downloadable.
- Executable from the OS of a mobile phone, smartphone, or digital tablet.

Q2. Solutions to minimize energy consumption (1.5 pts)

- Lightweight and flexible communication protocol.
- Use of internal memory for data requiring frequent transfer.
- Use of the Cloud for tasks with small data sizes and intensive computations.
- Use of interfaces (UI) with colors that consume less energy (e.g., dark background for OLED screens).

Q3. Diagram of the fundamental development steps (2 pts)

1. Definition → 2. Design → 3. Development → 4. Testing → 5. Deployment → 6. Tracking & User Feedback

Q4. Explain the 4th phase (Testing) (1.5 pts)

The testing phase is not mandatory but strongly recommended. It covers:

- **Functional tests:** Verify that features execute correctly.
- **Load tests:** Measure the server's capacity to handle multiple simultaneous connections (3G/4G).
- **Ergonomics tests:** Designed by ergonomists, carried out by representative users on targeted scenarios, to identify usability issues.

SUMMARY OF RECURRING COURSE QUESTIONS

Topic	Exams
Mobile app lifecycle (durable/transient states)	All exams
Advantages & disadvantages of Android	2021-2022, 2022-2023
Data storage locations	2022-2023, 2023-2024
Definition / criteria of a mobile app	2021-2022, 2024-2025 Mid-term
Definition of a service	2023-2024, 2024-2025 Makeup
Development lifecycle phases	2024-2025 Resit, Makeup, Mid-term
Characteristics of mobile terminals	2023-2024, 2024-2025 Makeup
Energy consumption	2024-2025 Mid-term